

Multiple regression

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One predictor is not enough

Holding something fixed

Is the model any good?

Do rich places vote Republican?

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Everyone is sure they know the answer.
Almost everyone is wrong.



- Every U.S. county that reported in 2024: 3,107 of them
- **trump_share** – percent of the county's presidential vote for Trump
- **median_income** – median household income (\$000)
- **pct_college** – percent of adults with a bachelor's degree or higher

Election returns from county-level 2024 returns; county characteristics from the Census ACS via OpenIntro.



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A tidy story. Money votes blue.

But richer counties are also better educated



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median income	-0.314 points per \$1,000	0.079
% college	-0.933 points per point	0.32

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So which one is actually doing the work?

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One predictor:

$$Y_i = \beta_0 + \beta_1(\text{income}) + \epsilon_i$$

Two predictors:

$$Y_i = \beta_0 + \beta_1(\text{income}) + \beta_2(\% \text{ college}) + \epsilon_i$$



One predictor:

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Same idea. Still least squares. We just choose three numbers instead of two.

```
model <- lm(trump ~ inc + ba, data=counties)
summary(model)
```

```
Call:
lm(formula = trump ~ inc + ba, data = counties)
```

```
Residuals:
```

	Min	1Q	Median	3Q	Max
	-61.967	-6.841	1.452	8.528	44.910

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	79.29733	0.90047	88.06	<2e-16 ***
inc	0.26630	0.02289	11.63	<2e-16 ***
ba	-1.21066	0.03376	-35.86	<2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 12.71 on 3104 degrees of freedom
```

```
Multiple R-squared:  0.3489, Adjusted R-squared:  0.3485
```

```
F-statistic: 831.6 on 2 and 3104 DF,  p-value: < 2.2e-16
```



$$\widehat{\text{trump}\%} = 79.3 + 0.266 (\text{income}) - 1.211 (\% \text{ college})$$

Alone, income was worth -0.314 points per \$1,000.

Next to education, it is worth $+0.266$.

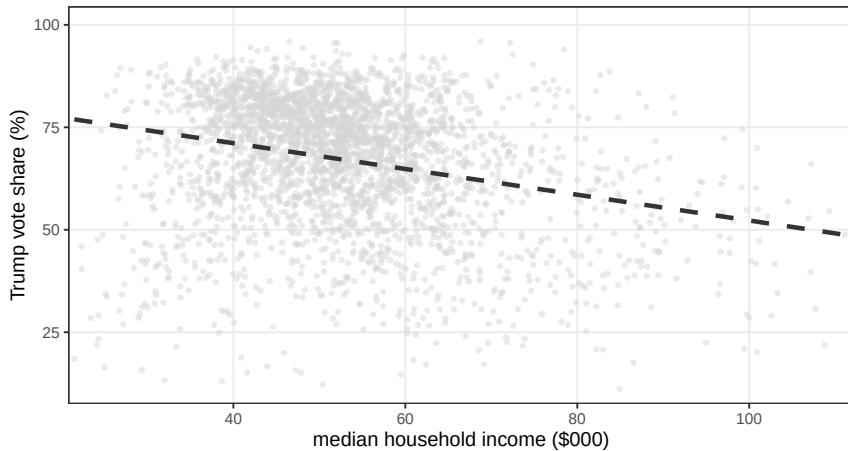


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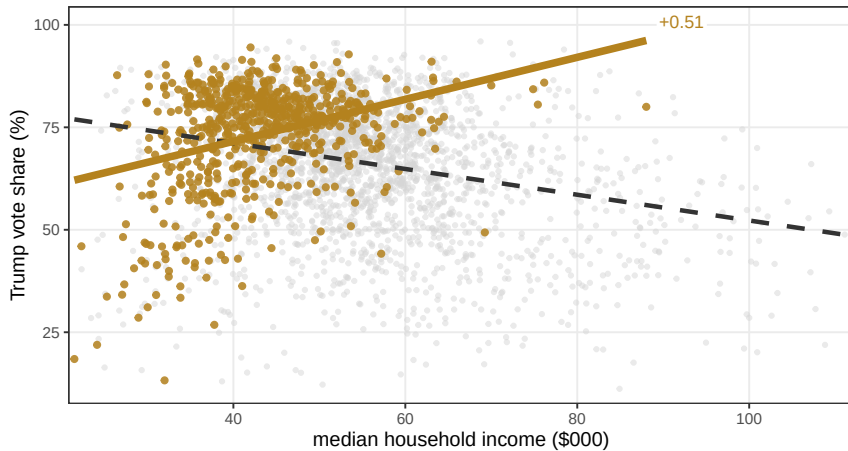
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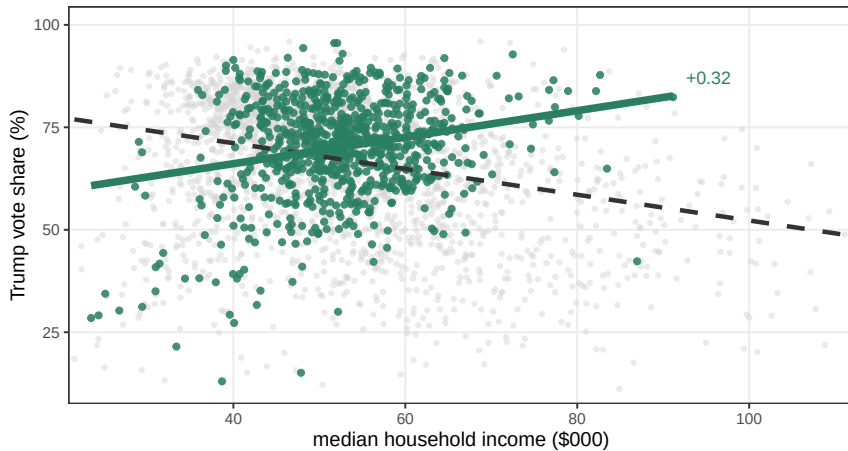
Same counties. Same election. What changed?



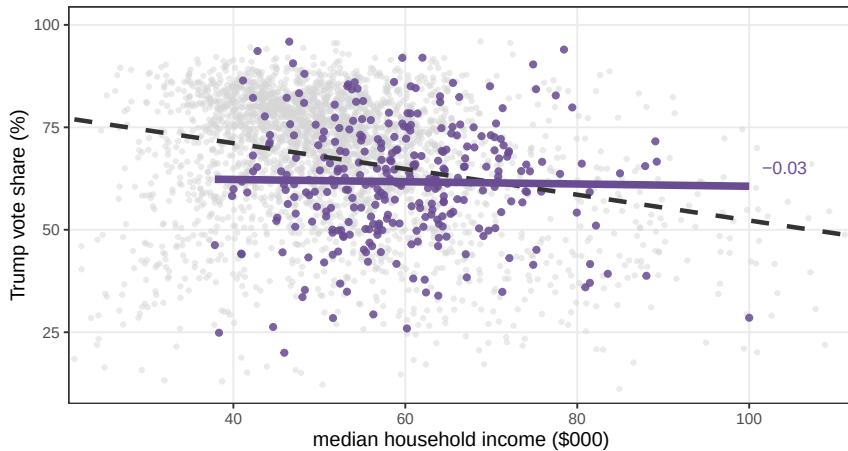
Only counties that are 10–15% college

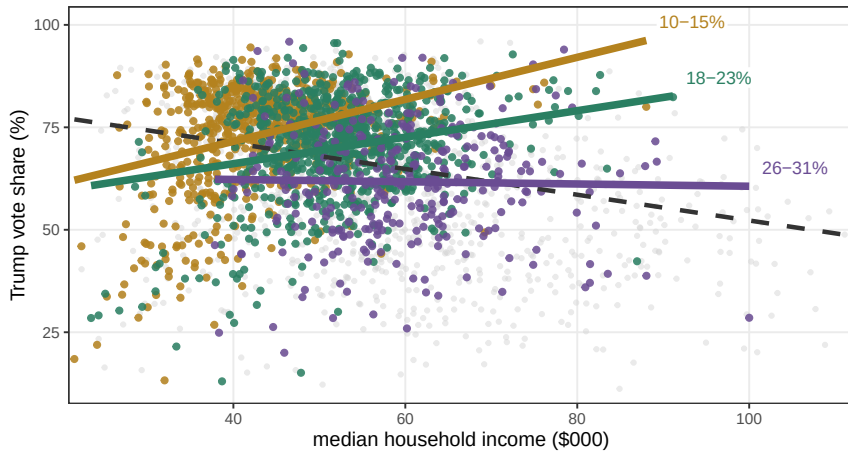


Only counties that are 18–23% college



Only counties that are 26–31% college







- The **dashed line** is the simple regression across all 3,107 counties. It slopes **down**.
- The **colored lines** hold education roughly fixed. They slope **up** (or flat).
- Compare two counties at different incomes in the raw data and you are also comparing a less-educated place to a more-educated one. You slide across the whole picture, not along one line.
- Hold education fixed and the sign reverses: $\hat{\beta}_1 = +0.266$. **Among similarly educated counties, richer ones voted more for Trump.**



- Income and education point in **opposite directions**. Because they travel together ($r = 0.71$), leaving education out lets income take the blame for it.
- Both slopes are correct. They answer **different questions**: “richer places” versus “richer places with the same education.”
- **A caution**: these are counties, not people. A pattern across counties need not hold for individual voters. That mistake has a name, the **ecological fallacy**, and it is everywhere in political commentary.

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Predictors	R^2	s_e
income	0.079	15.12
% college	0.32	12.99
both	0.349	12.71

- R^2 **always rises** when you add a variable, and the values **do not add**: $0.079 + 0.32 = 0.4$, but together they give only 0.349. The predictors overlap.
- $\sqrt{R^2} = 0.591$ is exactly the correlation between $\hat{Y}$ and Y .
- $s_e = 12.71$ points is the typical miss on one county. R also prints **adjusted** R^2 , which penalizes extra parameters.



- **Overall null:** $\beta_1 = \beta_2 = 0$, equivalently $R^2 = 0$ in the population, equivalently the model beats guessing $\bar{Y}$ by nothing. The F -test gives $F = 832$, $p < 2.2 \times 10^{-16}$. Rejected.
- **Individual coefficients:** test $H_0 : \beta_i = 0$ one at a time. This asks whether a predictor adds anything **once the other is in**. Income: $p = 1.2 \times 10^{-30}$. Education: $p = 6.6 \times 10^{-236}$.
- A smaller p -value does **not** mean a stronger predictor.



- Adding income to the education model is highly significant, yet R^2 moves only from 0.32 to 0.349.
- As a **prediction** tool it earns very little.
- But it completely reversed what we would say about money and voting. Sometimes a variable belongs in the model not to predict, but **to hold something fixed**.
- **There is no universal rule for adding or removing variables.**

A slope is never just a number.

It is an answer to the question
“compared to what?”